



NATIONAL UNIVERSITY OF MODERN LANGUAGES ISLAMABAD
FACULTY OF ENGINEERING & COMPUTING
DEPARTMENT OF COMPUTER SCIENCE
END-TERM EXAMINATION – FALL 2024 – BSCS VII A&B (MORNING)

Subject: Information Security
Allowed Time: 03 Hours

Instructor: Dr Zia ur Rehman
Total Marks: 50

Instructions: -

- L** Attempt all questions and return the question paper with the answer sheet.
- U** Clearly mention the question number before attempting it

QUESTION 1 [CLO-2, C3, PLO-3]

- ✓ Apply RSA algorithm to encrypt and decrypt after determining Public/Private PU/PR keys. Moreover, determine n , $\phi(n)$ as well for the following parameters. **[10 Marks]**

- $p = 7$; $q = 17$, $e = 11$; $M = 11$

QUESTION 2 [CLO-2, C3, PLO-3]

- ✓ Apply the ElGamal Digital Signature algorithm to sign the message "World" using the parameters:

- Prime number (p): 19, Primitive root (α): 13, Alice's Private Key (X_A): 7.
- Hash of the message (m): 16, Alice chooses $K = 5$.

Identify:

[10 Marks]

- Alice's Public Key (y), Signature values (S_1, S_2), Verify the signature to ensure its authenticity.

QUESTION 3 [CLO-3, C5, PLO-5]

✓ You are tasked with designing an access control system for a government database storing classified information. The database contains diplomatic cables and military intelligence. Your goal is to create a system incorporating role-based, rule-based, discretionary, and mandatory access control mechanisms. However, you need to plan a presentation for the higher authorities briefing about the types of access control and suggest the suitable type of access control for the following situations:

[10 Marks]

- For "Administrators", "System Analyst" and "Supervisors".
- For time-based, location-based, and behavior-based restrictions.
- Users control their data.
- Enforce strict access controls based on security classifications.

QUESTION 4 [CLO-3, C5, PLO-5]

- ✓ Determine critically and prioritize the main causes of Denial of Service (DoS) attacks in a large-scale online retail environment. **[10 Marks]**

QUESTION 5 [CLO-3, C5, PLO-5]

- ✓ Your organization wants to setup bastion host along with the packet filtering firewall to enhance security. The bastion host performs various functions, such as filtering traffic, monitoring network activity, and providing secure remote access to internal systems. **[10 Marks]**

- Assess the pros and cons of each bastion host type in the context of the given scenario, considering factors like cost, ease of setup and management, and suitability for their needs.